

Amritpal Singh Kaur

sharysingh1144@gmail.com | 07722 145 765 | London, UK | [apsk-dev.vercel.app](#) | [linkedin.com/in/amritpal-singh-kaur](#) | [github.com/Amrit004](#)

PROFESSIONAL SUMMARY

MSc Advanced Computer Science candidate (Queen Mary University of London, graduating 2026, expected Merit), currently working as an Onsite Software Support Specialist at Brock Solutions, Heathrow Airport, supporting real-time baggage operations across a multi-terminal enterprise network. Proficient in Python, JavaScript/TypeScript, AWS, Docker, and CI/CD pipelines, with a strong track record in security-hardened deployments, AI-integrated full-stack development, and formal software verification.

EDUCATION

MSc Advanced Computer Science, Queen Mary University of London

2025 - 2026

Expected Merit | Cloud Architecture, Machine Learning, Security & Authentication, Distributed Systems, Data Analytics, Research Project

- Final Year Project - Lightweight Software Verification Tool: designed and implemented a CTL model checker in Python achieving 34/34 test cases passing, covering Kripke structure parsing, temporal logic evaluation, and automated counterexample generation.

BSc (Hons) Computer Science, 2:1, Staffordshire University

2022 - 2025

Full-Stack Development, Cloud Computing, AI, Cybersecurity, Mobile Development, Databases, Networking, UX Design

PROFESSIONAL EXPERIENCE

Onsite Software Support Specialist, Brock Solutions

Apr 2026 - Present

Heathrow Airport, London

- Support SmartSuite, a real-time baggage operations platform used by major airlines, across a multi-terminal network of workstations, scanners, and peripherals.
- Serve as first point of contact for software incidents in a time-critical airport environment, diagnosing issues quickly to minimise operational impact.
- Collaborate with cross-functional teams to analyse, document, and recommend improvements to production systems.
- Support enterprise infrastructure including Windows Server, SQL databases, Azure cloud, and industrial automation interfaces.
- Contribute to the Service Team knowledge base through structured documentation and knowledge articles.

IT Support Engineer (Deployment / Break-Fix), Manpower (Zoom IT)

Apr 2022 - Apr 2026

Enterprise field placements at Bank of America, Amadeus, and ENI, London, UK

- Supported 100+ end users and resolved complex hardware/software incidents across enterprise desktop systems at Bank of America, Amadeus, and ENI, consistently minimising downtime in regulated FinTech, travel technology, and energy sector environments.
- Led large-scale deployment initiatives at Bank of America, completing 500+ device rollouts and asset management programmes, and delivered secure laptop/mobile device replacements with enterprise application installs for Amadeus travel technology operations.
- Provided Level 2 escalation support for complex technical incidents at Amadeus and assisted in a full Windows 11 migration across ENI's energy infrastructure systems, reducing mean time to resolution and maintaining 99.9% uptime.
- Implemented security patches, system hardening, and OS updates aligned with FinTech, GDPR, and financial industry regulatory requirements, while maintaining structured technical documentation and a team knowledge base across all three placements.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript (ES6+), TypeScript, C# (.NET), HTML5, CSS3

Frameworks & Libraries: React, Next.js, Node.js, Express.js, Scikit-learn, NumPy, Pandas, Matplotlib

Cloud & DevOps: AWS (EC2, S3, Lambda, IAM, CloudWatch), Azure (Azure AD, Intune), Docker, Kubernetes, CI/CD, Git, Vercel, Linux/Unix, Windows Server, Active Directory

Security: AES-256 Encryption, JWT Auth, OAuth, IAM, CVE Analysis, Network Security, Secure Design Patterns, FinTech/GDPR Compliance

Databases: SQL, MySQL, PostgreSQL, SQLite, SQL Server, MongoDB, Query Optimisation, ER Modelling

AI & Data: Machine Learning, NLP, Statistical Analysis, Predictive Modelling, Data Visualisation, Model Evaluation

Methodologies & Tools: Agile/Scrum, RESTful API Design, OOP, Functional Programming, Formal Verification, JIRA, Figma, Postman

KEY PROJECTS

- CTL Model Checker (Lightweight Verifier):** Pure-Python Computation Tree Logic model checker with Kripke structure support, 34/34 tests passing, counterexample generation, and lasso detection. Submitted as MSc research project.
- SOC Command Center:** Enterprise security operations dashboard with real-time threat detection, live attack visualisation, incident response workflows, and GDPR/SOC2/ISO27001 compliance tracking.
- CipherOS, Cryptography Toolkit:** Client-side browser toolkit implementing AES-256-GCM encryption, SHA-3/MD5 hashing, and JWT decoding via the Web Crypto API, with zero server-side dependencies.
- NetScan Pro, Vulnerability Scanner:** Nmap/Nessus-style scanner simulation with 4 scan modes and a CVE database of 10,000+ entries, including Log4Shell and EternalBlue.
- Enterprise Analytics Dashboard:** Next.js analytics platform with real-time KPIs, AI-powered revenue forecasting, and interactive Recharts visualisations.